AI Assisted Coding

Assignment – 16.3

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Task1:

Create schema and queries for a Hospital Management System.

Instructions:

• Tables: Doctors, Patients, Appointments.

• Use AI to define constraints (unique IDs, valid dates).

• Generate queries:

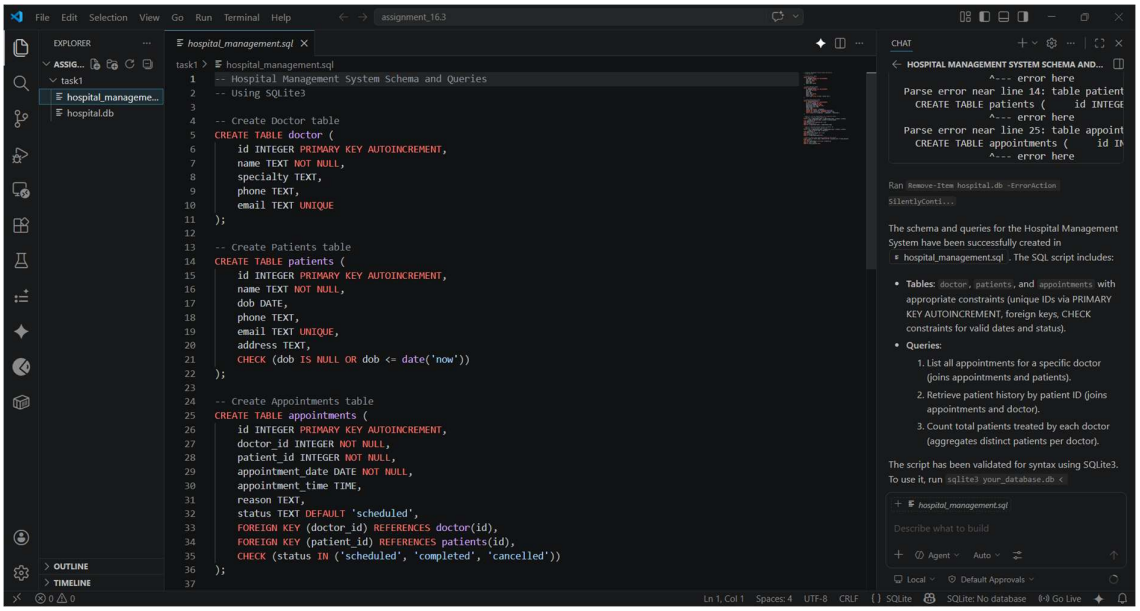
• List all appointments for a specific doctor.

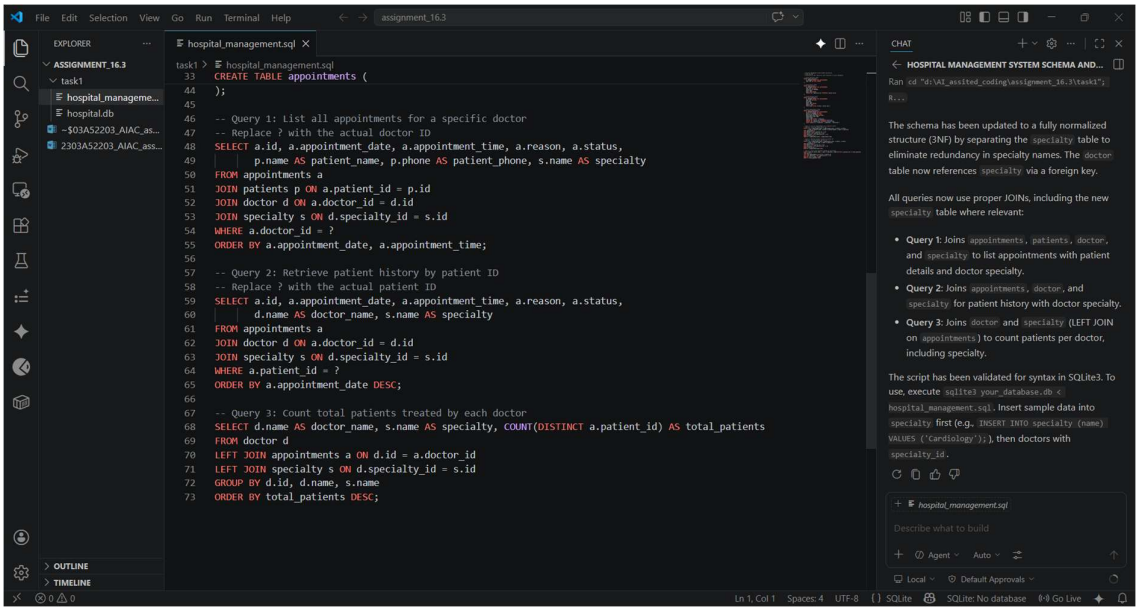
• Retrieve patient history by patient ID.

• Count total patients treated by each doctor.

Expected Output:

• Normalized schema and SQL queries with joins.





Task 2 – Real-Time Application: Online Shopping Database

Scenario:

Design a database for an E-commerce platform.

Requirements:

• Tables: Users, Products, Orders, OrderDetails.

• Generate queries to:

• Retrieve all orders by a user.

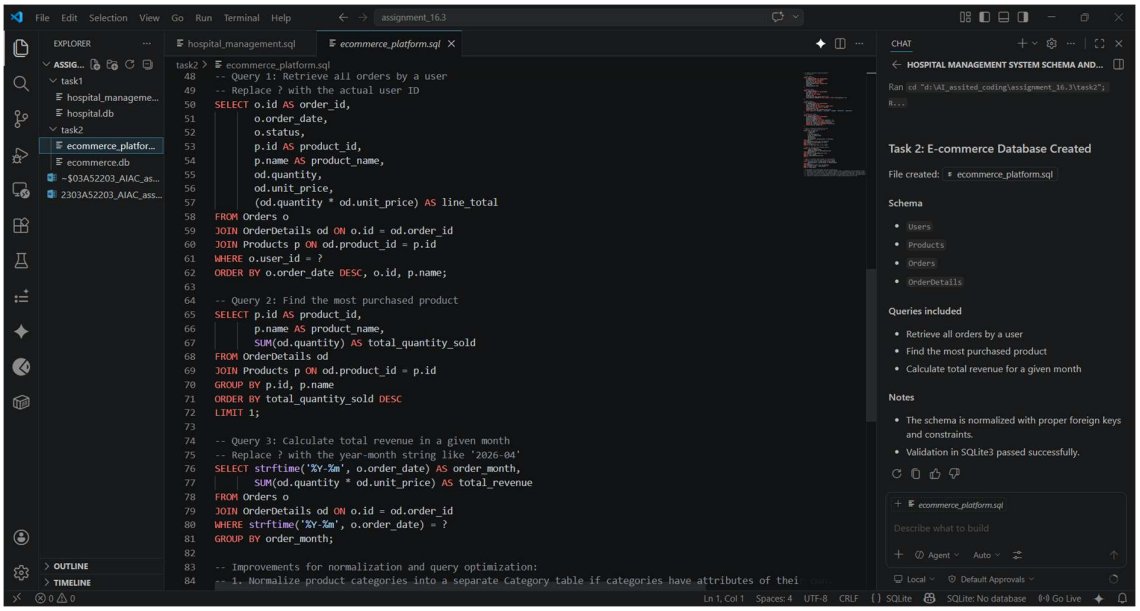
• Find the most purchased product.

• Calculate total revenue in a given month.

• AI should also suggest normalization improvements and query optimization.

Expected Output:

• Complete schema with relationships + SQL queries for analytics.



Task 3 – Library Database

Task:

Design schema for a Library Management System.

Instructions:

• Tables: Books, Members, Loans.

• Use AI to suggest indexing strategy for faster queries.

• Generate queries:

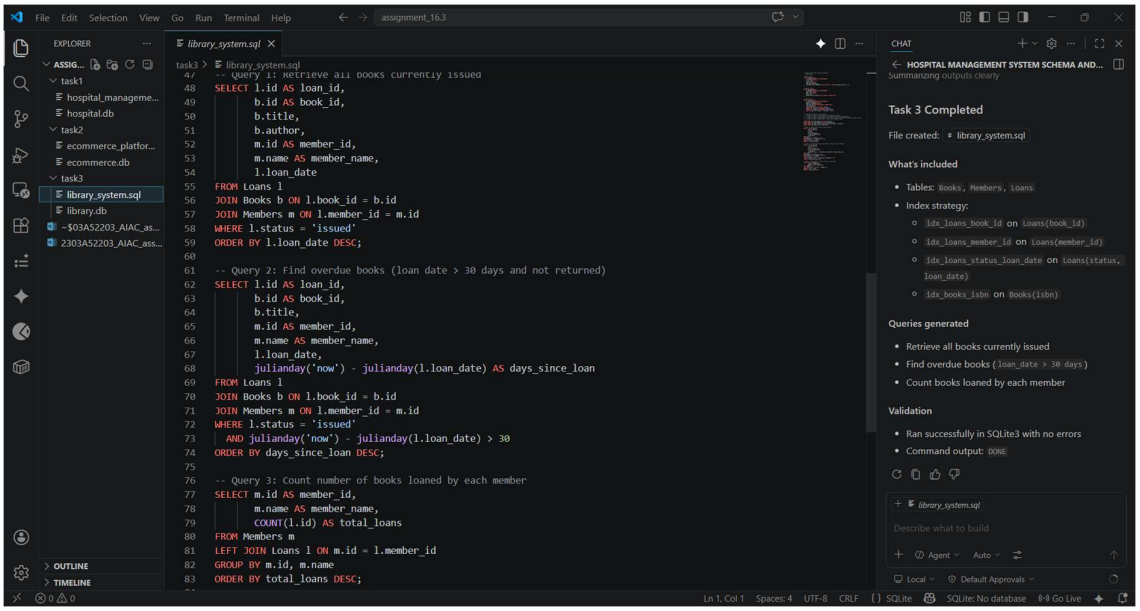
• Retrieve all books currently issued.

• Find overdue books (loan date > 30 days).

• Count number of books loaned by each member.

Expected Output:

• Schema + SQL queries demonstrating joins and conditions.


File Edit Selection View Go Run Terminal Help
assignment_16.3
EXPLORER
ASSIG...
task1
hospital_manageme...
hospital.db
task2
ecommerce_platfor...
ecommerce.db
task3
library_system.sql
library.db
~$03A52203_AIAC_as...
2303A52203_AIAC_ass...
OUTLINE
TIMELINE
library_system.sql
task3 > library_system.sql
-- Query 1: Retrieve all books currently issued
SELECT l.id AS loan_id,
b.id AS book_id,
b.title,
b.author,
m.id AS member_id,
m.name AS member_name,
l.loan_date
FROM Loans l
JOIN Books b ON l.book_id = b.id
JOIN Members m ON l.member_id = m.id
WHERE l.status = 'issued'
ORDER BY l.loan_date DESC;
-- Query 2: Find overdue books (loan date > 30 days and not returned)
SELECT l.id AS loan_id,
b.id AS book_id,
b.title,
m.id AS member_id,
m.name AS member_name,
l.loan_date,
julianday('now') - julianday(l.loan_date) AS days_since_loan
FROM Loans l
JOIN Books b ON l.book_id = b.id
JOIN Members m ON l.member_id = m.id
WHERE l.status = 'issued'
AND julianday('now') - julianday(l.loan_date) > 30
ORDER BY days_since_loan DESC;
-- Query 3: Count number of books loaned by each member
SELECT m.id AS member_id,
m.name AS member_name,
COUNT(l.id) AS total_loans
FROM Members m
LEFT JOIN Loans l ON m.id = l.member_id
GROUP BY m.id, m.name
ORDER BY total_loans DESC;
CHAT
HOSPITAL MANAGEMENT SYSTEM SCHEMA AND...
Summarizing outputs clearly
Task 3 Completed
File created: library_system.sql
What's included
Tables: Books, Members, Loans
Index strategy:
idx_loans_book_id on Loans(book_id)
idx_loans_member_id on Loans(member_id)
idx_loans_status_loan_date on Loans(status, loan_date)
idx_books_isbn on Books(isbn)
Queries generated
Retrieve all books currently issued
Find overdue books (loan_date > 30 days)
Count books loaned by each member
Validation
Ran successfully in SQLite3 with no errors
Command output: DONE
library_system.sql
Describe what to build
Agent
Auto
Local
Default Approvals
Ln 1, Col 1
Spaces: 4
UTF-8
CRLF
SQLite
SQLite: No database
Go Live